

Finite Commutative Rings

We have examined the structure of finite fields. What can we say about the structure of finite commutative rings? Note that if R is an order, and n a positive integer, then $R/\langle n \rangle$ is such a ring. So this question is of some interest in number theory.

We will make use of some basic results about primary ideals in commutative rings that are proved elsewhere.

Note that every ideal in a finite ring is finite and thus finitely generated. Thus a finite ring is Noetherian.

Prime ideals are maximal

If D is a finite (commutative) domain, then for any non-zero $a \in D$, the map $D \rightarrow D$ given by $x \mapsto ax$ is a one-to-one map. Since D is finite, this map is onto as well. It follows that D has division by non-zero elements. Hence D is a field.

(**Note:** One need not assume that D is commutative, as one can prove that a finite domain is always commutative.)

Now, suppose that R is a finite ring and P a prime ideal in R . Then R/P is a finite domain, hence a field. This shows that P is a maximal ideal.

In a finite ring every prime ideal is maximal.

Note that since R is finite, it has only finitely many subsets, hence only finitely many ideals. It follows that it has only finitely many maximal ideals.

Nilpotent ideal

Given a collection of $\{M_1, \dots, M_k\}$ of maximal ideals in a commutative ring and any positive integers $n_1, \dots, n_k$ we have $M_1^{n_1} \cap \dots \cap M_k^{n_k} = M_1^{n_1} \dots M_k^{n_k}$.

In a finite ring any prime ideal is maximal and the collection of all maximal ideals $\{M_1, \dots, M_k\}$ is a finite set.

Since the intersection of all prime ideals is the ideal of all nilpotent elements, we see that there is an n such that $(M_1 \dots M_k)^n$ is the zero ideal. It follows that $M_1^n \dots M_k^n = 0$. Thus, we see that the map

$$R \rightarrow R/M_1^n \times \dots \times R/M_k^n$$

is injective for n sufficiently large.

Recall that a ring is said to be a *local ring*, if it has a unique maximal ideal M . It follows that every element of $R^* = R \setminus M$ is a unit. Conversely, if R^* is the set of units in R and $M = R \setminus R^*$ is an ideal, then it is a maximal ideal and R is a local ring.

In particular, if M is a maximal ideal in a ring R , then R/M^n is a local ring.

Descending sequences

Given a maximal ideal M in a finite ring R consider the (non-increasing) sequence of ideals

$$M \supset M^2 \supset M^3 \subset \dots$$

Since R has finitely many subsets, we must have $M^n = M^{n+1}$ for some $n > 0$. It easily follows that $M^n = M^{n+k}$ for all $k > 0$. In other words, the above sequence is finite.

We denote by n_M be the positive integer associated in this way to a maximal ideal M in the finite ring R .

Chinese Remainder Theorem

Given ideals I and J in R such that $I + J = R$, there are elements $a \in I$ and $b \in J$ such that $1 = a + b$. Now, given c and d in R , we consider the element $e = cb + da$ in R and note that

$$\begin{aligned} e &= cb + da = c - ca + da \in c + I \\ e &= cb + da = d - db + cb \in d + J \end{aligned}$$

which shows that the natural homomorphism $R \rightarrow R/I \times R/J$ is onto.

Given a collection of $\{P_1, \dots, P_k\}$ of prime ideals in a commutative ring R and a maximal ideal M that does not contain any of the P_i . For any positive integers $n, n_1, \dots, n_k$ we can show that there is an $a \in M^n$ and $b \in P_1^{n_1} \dots P_k^{n_k}$ such that $1 = a + b$; equivalently $R = M^n + P_1^{n_1} \dots P_k^{n_k}$.

Applying the above inductively to a set of maximal ideals $\{M_1, \dots, M_k\}$ and positive integers $n_1, \dots, n_k$, we see that

$$R \rightarrow R/M_1^{n_1} \times \dots \times R/M_k^{n_k}$$

is surjective for all n .

Decomposition

Combining the above results, we see that for a finite ring R :

- there is a finite set of prime ideals $\{M_1, \dots, M_k\}$ which are all maximal ideals;
- for each $i = 1, \dots, k$ there is a positive integer n_i such that $M_i^{n_i} = M_i^{n_i+1}$;
- the map

$$R \rightarrow R/M_1^{n_1} \times \dots \times R/M_k^{n_k}$$

is an isomorphism.

Decomposition

Combining the above results, we see that for a finite ring R :

- there is a finite set of prime ideals $\{M_1, \dots, M_k\}$ which are all maximal ideals;
- for each $i = 1, \dots, k$ there is a positive integer n_i such that $M_i^{n_i} = M_i^{n_i+1}$;
- the map